

IBM CAPSTONE TODAY AND FUTURE OF CODING

August, 2026

INSIGHTS

1. Lab-2

- Washington DC is the province with the largest quantity of available jobs on the market. Philadelphia is with the lowest amount in that regard.

2. Lab-4

- Swift is the software language that yields the most annual salary to its experts. It is followed by Python, C++ and JavaScript. On the other hand, PHP is the one that provides the lowest amount in this regard.

3. Lab-5

- Mean age of all coders from 185 countries is around 33.

4. Lab-6

- More than one-fifth of all coders are full-stack developer by profession, full-time employed and work in hybrid mode: both in-person and remote. This definition is the major blanket for the most countries.
- In USA, however, the most preferred work mode is remote.

5. Lab-8

- More than half of the coders participating in the survey did not provide an answer for the questions about "Job Satisfaction" and "Yearly Compensation" as well as AI related ones.

6. Lab-10

- Yearly Compensation data is right-skewed, but there is a peak of nearly 2500 respondents with 10000 dollars or less annual income. Nearly 80% of those are between 18-34 years old.

7. Lab-12

- Job satisfaction average among respondents IAW work experience do not vary much, up until the experience in coding exceeds 20 years.
- Job satisfaction is around 7 over 10 among respondents with less than 20 years of experience, while it is 8 over 10 for the remaining. But the remaining, with more than 20 years of experience in coding, accounts only for the 15% percent of respondents.
- However, this renders the general job satisfaction mean around 8.

8. Lab-13

- More than one quarter of respondents have 5 years or less in coding.
- JavaScript is the language that nearly 60% of the respondents indicated that they knew to code in. It is followed by HTML/CSS and Python.
- Python is the language that nearly 40% of the respondents indicated that they want to learn coding in. It is followed by JavaScript and SQL.
- Freelancers or independent contractors are mainly secondary school graduates.
- Full-time employees are mainly bachelor's degree holders.

9. Lab-14

- Nearly one fifth of respondents indicated that they work in software development industry, while some less than 1 percent indicated that they work in insurance or energy sectors.

10. Lab-15

- The correlation between the yearly annual compensation and job satisfaction is positive but moderate.

11. Lab-16

- More than 35% of respondents are between 35 and 44 years old.
- 75% of respondents reported a total compensation of 100000 at the least.
- Most used database is PostgreSQL with the use of 34% of all respondents. It is followed by SQLite with nearly 20%.

12. Lab-17

- More than half of respondents with 500000 dollars or more annual income are 25-34 years old.
- Respondents with less than 8 years of experience in coding are the ones that provided the most and the highest metrics in job satisfaction.

13.Lab-18

- The highest and stretched yearly compensation comes from the respondents who do not provide their age, EXCLUDING outliers.
- Yearly compensation distribution is similar for back-end, full-stack and desktop developers. But it is 20% lower for the front-end developers.
- India is the country whose respondents reported by far the most stretched annual income range EXCLUDING outliers, among the five countries that majority of respondents are from.

14.Lab-19

- Job satisfaction average is at its highest (slightly more than 25 over 100) for the respondents of 18-34 years old.
- Job satisfaction is at its low ebb (around 15%) for respondents who indicated Julia and at its peak (around 28%) for respondents who indicated Solidity as languages to work with.
- Job satisfaction is its low ebb of 11% for the retired respondents, -and even less for the respondents who receives around 100000 annual compensations in general- and strangely at its highest of 32% for the unemployed respondents.

15.Lab-20

- Interestingly, respondents of less than 18 and more than 65 years old reported the most frequent participation in the survey: multiple times per day.
- Most of respondents are 18-44 years old, 24-34 years old to be more specific, and with the knowledge of Python, JavaScript, HTML/CSS, SQL and Typescript.
- Respondents with the most job satisfaction levels use Raven DB database and/or Gatsby web frame, while some others with the least job satisfaction levels use Couch DB database and/or Django web frame.
- Yearly compensation is the highest for engineering managers and the lowest for academic researchers EXCLUDING students.
- For respondents of age 18-44, the most preferred collaboration tools used are Visual Studio Code, Visual Studio and Vim.
- The most used web frames are React, node.js and next.js.

16.Lab-21

- The most popular operating system is Windows with the 35% of respondents reported use.
- The most (29%) of respondents use AI tools for writing code and some 12% use it for creating synthetic data.
- The most used embedded is Raspberry Pi with 31% of respondents reported use.

17.Lab-22

- The most popular Platforms are Amazon Web Services, Microsoft Azure and Google Cloud

18. Lab-23

- Total compensation seems to be almost linearly and positively correlated with age and work experience, except that the increase halts and reverts to decrease after the age of 65.
- The job satisfaction gradually decreases for the respondents having a work experience of 3 years or more, but with tiny fluctuation up until 32 years of experience. After that, job satisfaction averages fluctuate considerably between age increments.
- As with the job satisfaction, annual compensation also fluctuates considerably for respondents with more than 32 years of work experience.

19. Lab-24

- USA is the country where majority of respondents are from, around 15% percent.

Name	Title	Date
Ali Can TURAN	Associate Data Analyst	August 2026